NORTH CAROLINA AGRICULTURAL AND TECHNICAL STATE UNIVERSITY

MATH 224-Data Analysis Project

Part III: Regression Modeling & The Whole Story

AI Tool Usage Instructions

For this project, you are allowed to use one generative AI tool of your choice. Examples include ChatGPT, Microsoft Copilot, or Google Gemini. Our school currently provides free access to Microsoft Copilot, and Gemini offers a free student plan for 12 months — but it is completely up to you which tool you prefer.

Rules for AI Use

- Use only ONE dedicated conversation/chat for this project. All project-related prompts and questions must stay in that same conversation.
- Do not spread project-related questions across multiple chats. If you need help for unrelated study purposes, open a different chat.
- Include your AI interaction link upon submission so your instructor can review your full conversation.
- AI is a tool to assist your thinking — not a replacement for it. You must understand, verify, and take ownership of every answer you submit.

What You SHOULD Do ✓

✓ Good AI Use — Examples

"I have a dataset with variables age, income, and education. Can you explain what summary statistics I should compute and why?"

"My histogram looks right-skewed. What does that mean for interpreting the mean vs. median?"

"Here is my R code — it gives an error. Can you help me find the issue?" [paste your actual code]

"Can you explain what a confidence interval means in simple terms?"

"I got a p-value of 0.03 in my t-test. How do I interpret this in context?"

What You Should NOT Do ✗

✗ Prohibited AI Use — Do NOT do these

Do NOT paste the project question directly and submit whatever the AI writes.
Do NOT ask: "Do Part 3 of my MATH 224 project for me."
Do NOT copy and paste AI output word-for-word without understanding it.
Do NOT use AI to generate R code you cannot explain or did not verify.
Do NOT use multiple chat threads to hide the extent of your AI use.

⚠ Important Reminder

AI tools can make mistakes, especially with statistics and R code. Always verify outputs against your course notes and textbook. If AI gives you something that seems off, trust your judgment and ask your instructor.

Reflection Requirement

At the end of your submission, include a short paragraph (5–8 sentences) reflecting on how you used the AI tool. Address the following:

- Which AI tool did you use and for which tasks?
- Did the AI make any mistakes? How did you catch and correct them?
- How did using AI affect your understanding of the material?
- What would you do differently next time?

General Instructions

Use the R Markdown report outline template you created in RStudio Cloud during lab to write a final report addressing the tasks below. Knit your code as a PDF and submit it — including all code and answers — under "Data Analysis Project – Part III" in Blackboard by the due date.

In Parts I and II, you explored your dataset, identified research questions, performed exploratory data analysis, and used statistical inference. In this final part, you will build regression models and compile your work into a complete story-telling report.

Project Tasks

1 — Correlation and Regression Modeling (25 points)

1a. Scatter Plots & Correlation Coefficients (6 points)

Identify all numerical (quantitative) variables in the dataset. Use select() to select only the response variable and numerical explanatory variables. Then use ggpairs() from the GGally package (install it first) to create scatter plots and compute correlation coefficients.

In your write-up, describe the associations between pairs of variables based on the scatter plots. Which pair has the strongest linear association?

AI Prompt Ideas for 1a

"I used ggpairs() and got this output: [describe]. Can you help me describe the associations I see in the scatter plot matrix?"

"How do I install and load the GGally package in R?"

✗ Do NOT do this

Do NOT ask AI to interpret your scatter plots for you without first attempting your own description.

Do NOT report correlations without contextualizing them in your dataset.

1b. Multiple Linear Regression Model (6 points)

Develop a multiple linear regression model to predict your response variable using all relevant explanatory variables. Write the model equation using variable names from your dataset. Use lm() to estimate the model and report the output using summary().

AI Prompt Ideas for 1b

"I am building a multiple regression model in R to predict house price. My explanatory variables are lot size, number of bedrooms, and age. Can you show me the lm() syntax?"

"Here is my summary() output: [paste]. Can you help me understand what the F-statistic and R-squared values mean?"

✗ Do NOT do this

Do NOT ask AI to write your complete regression model without running or understanding it yourself.

Do NOT copy an AI-generated model equation without verifying the variable names match your actual dataset.

1c. Interpret the Regression Equation (5 points)

Write the estimated regression equation using the actual coefficient values from your summary() output. Then interpret the results, describe how each explanatory variable affects or correlates with the outcome variable.

AI Prompt Ideas for 1c

"My regression output gives a coefficient of -0.34 for City MPG in predicting car price. Help me write a sentence interpreting this coefficient in context."

"One of my coefficients is not statistically significant (p-value =). What does this mean and should I remove the variable?"

1d. Model Quality (4 points)

Describe the quality of your model by reporting and interpreting the R-squared value. What percentage of the total variation in the response variable is explained by the explanatory variables collectively? Is the overall regression model statistically significant?

AI Prompt Ideas for 1d

"My model has an R-squared of 0.68. Can you help me explain what this means for my dataset about [topic]?"

"My F-statistic p-value is 0.0001. What does this mean about the overall significance of my regression model?"

1e. Residual Plots (4 points)

Use the plot() function on your fitted model to produce residual plots. Comment on each plot. Do the residuals suggest that all regression assumptions are satisfied (linearity, constant variance, normality)?

AI Prompt Ideas for 1e

"My Residuals vs Fitted plot shows a curved pattern. What does this suggest about my regression assumptions?"

"My Q-Q plot has points deviating at the tails. Is this a major concern for my regression analysis?"

✗ Do NOT do this

Do NOT ask AI to describe your residual plots for you without first looking at them yourself.
Do NOT report that all assumptions are satisfied without actually examining the plots.

2 — Final Story-Telling Report (15 points)

Compile all your work from Parts I, II, and III above into a single, coherent final report using your R Markdown template. The report should read as a complete story of your data analysis, not just a list of answers.

Your final report must include:

- An introduction that provides context for your dataset and research questions
- Exploratory data analysis from Part I (summary statistics and graphs)
- Statistical inference from Part II (confidence intervals and interpretations)
- Regression analysis of Part III
- A conclusion that ties together your findings and answers your original research questions
- Your AI reflection paragraph

AI Prompt Ideas for Task 2

"I have completed my analysis. Here are my main findings: [summarize]. Can you help me write a conclusion paragraph that ties these findings together?"

"Can you review my introduction paragraph and suggest how to make it clearer or more engaging? Here it is: [paste]"

"My report feels disjointed. Here are my three sections: [paste]. How can I add transitions to make it read as one story?"

✗ Do NOT do this

Do NOT ask AI to write the entire report for you.
Do NOT submit an AI-generated report without personalizing it with your actual results and interpretation.
Do NOT ignore the template structure provided in Blackboard.

Submission Checklist

- R Markdown knitted as a PDF with all code visible
- Copy and paste the entire AI conversation, in addition to the link, into a text doc (.txt)